

SM PHARMACEUTICALS SDN BHD
Polynex (Sodium Polystyrene Sulfonate) Powder

DESCRIPTION:

A cream to light brown, odourless and tasteless fine powder.

COMPOSITION:

Sodium Polystyrene Sulfonate Powder 500 gm

PHARMACOLOGY (SUMMARY OF PHARMACODYNAMICS AND PHARMACOKINETICS):

Antihyperkalemic – In the intestine (mostly the large intestine), sodium ions are released and are replaced by potassium and other cations before the resin is passed from the body.

Sodium polystyrene sulfonate also exchanges small amounts of other cations such as magnesium and calcium.

The *in vitro* exchange capacity is approximately 3.1 mEq (in vivo 1 mEq) / g of potassium of Polynex.

Absorption:

Not absorbed from gastrointestinal tract; exchanges sodium for potassium and is excreted from the intestine.

Onset of action:

Hours to days: therefore, other measures such as dialysis may be necessary in emergency situations.

INDICATIONS:

It is indicated in the treatment of hyperkalemia associated with severe oliguria or anuria due to acute renal failure, e.g., septic abortion, complicated labour, incompatible blood transfusion, crush injury, prostatectomy, severe burns, surgical shock and in renal failure, as in some cases of glomerulonephritis and severe pyelonephritis. Hyperkalaemia in patients requiring haemodialysis or on prolonged peritoneal dialysis.

CONTRAINDICATIONS:

Risk benefit should be considered when the following medical problem exist:

Edematous conditions, such as: severe congestive heart failure or severe hypertension (may require compensatory restriction of sodium intake from other sources or use of dialysis instead of sodium polystyrene sulfonate)

Sensitivity to sodium polystyrene sulfonate.

Sodium polystyrene sulphonate should not be administered orally to neonates, and is contraindicated by any route in neonates with reduced gut motility or in any patient with obstructive bowel disease. Avoid calcium-containing resin in hyperparathyroidism, multiple myeloma, sarcoidosis, or metastatic carcinoma.

Administration should be stopped when the serum potassium level falls to 5mmol/L. the product also cannot be used in patients with obstructive bowel disease.

SIDE EFFECTS / ADVERSE REACTIONS:

Anorexia, nausea, vomiting, constipation, and occasionally diarrhoea may develop during treatment with sodium polystyrene sulphonate. Constipation may be severe; large doses in elderly patients and in children may result in faecal impaction and gastro-intestinal concretions have occurred following oral administration to neonates.

If necessary a mild laxatives such as sorbitol may be used to prevent or treat constipation; magnesium-containing laxatives should not be used.

Serious potassium deficiency can occur with sodium polystyrene sulphonate and signs of severe hypokalaemia may include irritability, confusion, ECG abnormalities, cardiac arrhythmias, and severe muscle weakness. Like other cation-exchange resins, sodium polystyrene sulphonate is not totally selective and its use may result in other electrolyte disturbances such as hypocalcaemia. Significant sodium retention may also occur, especially in patients with impaired renal function, and may lead to congestive heart failure.

Those indicating need for medical attention

Incidence less frequent – dose related

Fecal impaction (severe stomach pain with nausea and vomiting); **hypocalcemia** (abdominal and muscle cramps); **hypokalemia** (confusion with irritability; delayed thought processed; irregular heartbeat; severe muscle weakness); **sodium retention** (decrease in urination, swelling of hands, feet or lower legs; weight gain).

Those indication need for medical attention only if they continue or are bothersome

Incidence more frequent – especially with large doses

Constipation, loss of appetite, nausea or vomiting

PRECAUTIONS / WARNINGS:

Carcinogenicity / Mutagenicity:

Studies have not been done to evaluate the carcinogenic or mutagenic potential of sodium polystyrene sulfonate.

Care is needed with rectal administration to neonates and children in order to avoid impaction of the resin. Treatment should be discontinued if clinically significant constipation develops. Sorbitol has been recommended for the prophylaxis and treatment of constipation, magnesium-containing laxatives should not be used.

For constipation, sorbitol (10-20 mL of 70% syrup, every 2 hours as needed until 1 or 2 watery stools daily are produced) should be used. This measure also reduces any tendency to fecal impaction.

Patients receiving sodium polystyrene sulphonate should be monitored for electrolyte disturbances, especially hypokalaemia. Since serum concentrations may not always reflect intracellular potassium deficiency, symptoms of hypokalaemia should also be watched for and the decision to stop treatment assessed individually. Hypokalaemia may exacerbate the adverse effects of digoxin and sodium polystyrene sulphonate should be used with caution in patients receiving cardiac glycosides.

Administration of sodium polystyrene sulphonate can result in sodium overloading and it should be used cautiously in patients with renal failure and with conditions requiring a restricted sodium intake, such as congestive heart failure and severe hypertension.

Metabolic alkalosis has been reported in patients, particularly those with renal disease, after the concomitant oral administration of sodium polystyrene sulphonate and cation-donating antacids

and laxatives such as magnesium hydroxide, aluminium hydroxide, or calcium carbonate; the potassium-lowering effect will also be diminished.

The possible effects of sodium polystyrene sulphonate on serum electrolytes should be considered when diagnostic measurements are contemplated in patients receiving such treatment. Following the administration of sodium polystyrene sulphonate retention enemas, the colon should be irrigated to ensure removal of the resin.

Paediatrics:

Appropriate studies on the relationship of age to the effects of sodium polystyrene sulfonate have not been performed in the paediatric population. However, paediatric-specific problems that would limit the usefulness of this medication in children are not expected.

Geriatrics:

Although appropriate studies on the relationship of age to the effects of sodium polystyrene sulfonate have not been performed in the geriatric population, the elderly may be more likely to develop fecal impaction.

Pregnancy / Reproduction:

Pregnancy – Studies in humans have not been done in humans and animals.

Breast-feeding:

It is not known whether sodium polystyrene sulfonate is distributed into breast milk.

DRUG INTERACTIONS

Antacids or Laxatives:

(Sodium polystyrene sulfonate may bind with magnesium or calcium found in nonsystemic antacids or laxatives, preventing neutralization of bicarbonate ions and leading to systemic alkalosis that may be severe; concurrent use is not recommended, although the risk may be more less with rectal administration of the resin)\

Diuretics, potassium-sparing or Potassium supplements:

(Sodium polystyrene sulfonate reduces serum potassium concentrations by replacing potassium with sodium; fluid retention may occur in some patients because of the increased sodium intake).

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:

For oral or rectal administration only. The suspension should be freshly prepared and used within 24 hours. The precise dose requirements should be decided on the basis of frequent serum electrolyte determinations. Heating may alter the exchange properties of the resin.

This product should not be taken with fruit squash, which contains high levels of potassium.

Usual adult dose:

Antihyperkalemic :

Oral, 15 grams (4 level teaspoonfuls) one to four times a day, up to 40 grams four times a day.

The resin is given by mouth in a little water or it may be made into a paste with some sweetened vehicle.

Usual paediatric dose:

Antihyperkalemic :

Oral, usually 1 gram per kg of body weight per dose as needed to correct hyperkalemia.

Dosage may be reduced to 0.5g/kg body weight in divided doses for maintenance therapy.

For Rectal Administration:

When oral administration is not possible, the resin may be given rectally as a suspension of 30-50g resin in 100 mL 2% methylcellulose and 100 mL water as a daily retention enema. At the beginning of treatment administration rectally as well as orally may help to achieve a more rapid lowering of the serum potassium.

The enema of Polynex should be retained for several hours (at least 9) and be followed by a non-sodium cleansing enema.

Children: 1g/kg body weight daily in 1-4 doses in acute hyperkalaemia. Dosage may be reduced to 0.5g/kg body weight in divided doses for maintenance therapy.

Preparation of dosage form:

To facilitate and hasten action and prevent constipation, Sodium Polystyrene Sulfonate should be suspended in 3 to 4 mL of 70% sorbitol syrup (which acts as an osmotic cathartic) per gram of resin. The resin also may be mixed with water or a diet appropriate for a patient with renal failure and administered orally through a plastic tube.

For rectal administration, Sodium Polystyrene Sulfonate suspended in 100 to 200 mL of an aqueous vehicle (for example, 25% sorbitol, 1% methylcellulose, or 10% dextrose). Care should be taken that the paste is not too thick because it will be less effective.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE AND ANTIDOTE(S):

In the event of overdosage, the resin should be removed from the alimentary tract by the use of laxatives or enemas. Appropriate measures should be taken to restore serum potassium levels to normal.

PACKING / PACK SIZES:

500 gm in HDPE container

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

Store below 30°C, in a well-closed container. Shake well before use.

Shelf Life: 3 years from the date of the manufacture.

NAME AND ADDRESS OF MANUFACTURER AND DISTRIBUTOR IN MALAYSIA:

SM PHARMACEUTICALS SDN BHD (218620-M)

Lot 88, Sungai Petani Industrial Estate

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